



Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + SVM + Decision Tree

**MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA**

**Inteligência Artificial para as Telecomunicações
2025/2026**

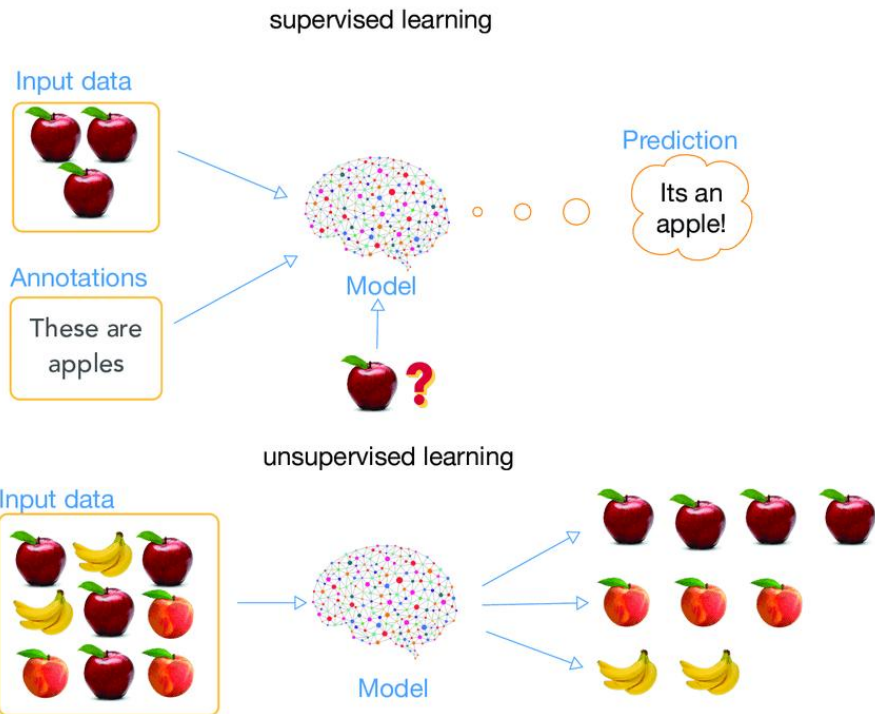


Objectives

- The primary goal of this task is **to develop models capable of clustering wines based on relevant characteristics**, such as acidity, sugar content, pH, alcohol level, etc.

Machine Learning

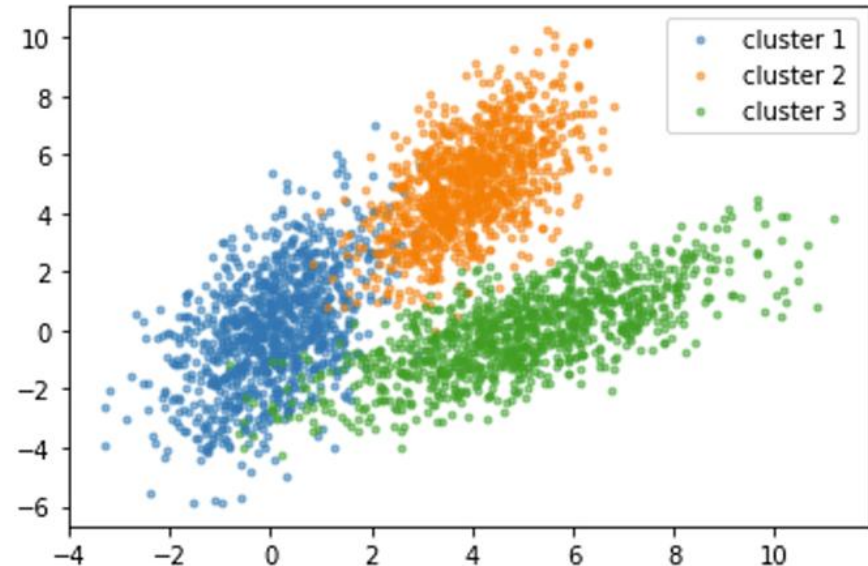
- **Supervised learning:** uses labeled data to learn the correct outputs (e.g., predicting a car's price from its features).
- **Unsupervised learning:** uses completely unlabeled data; the algorithm searches for hidden patterns on its own.



Clustering

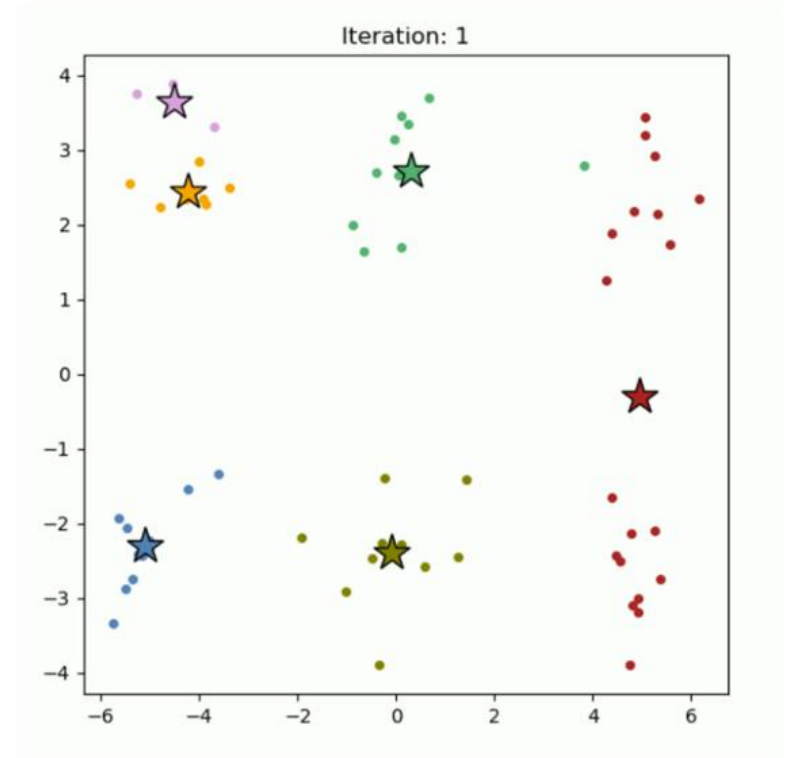
Clustering is a Machine Learning technique that **groups objects**, or data points, **based on similarities**.

A cluster = a set of similar points.



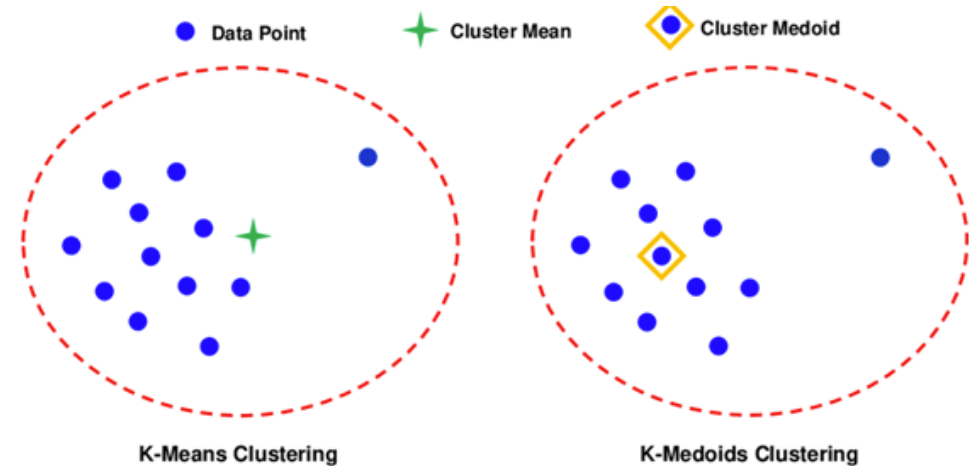
Clustering - K-Means

K-means is a widely used unsupervised clustering algorithm that partitions data into a **preset number of groups (k)**. It relies on centroids—the average position of the points in each cluster.



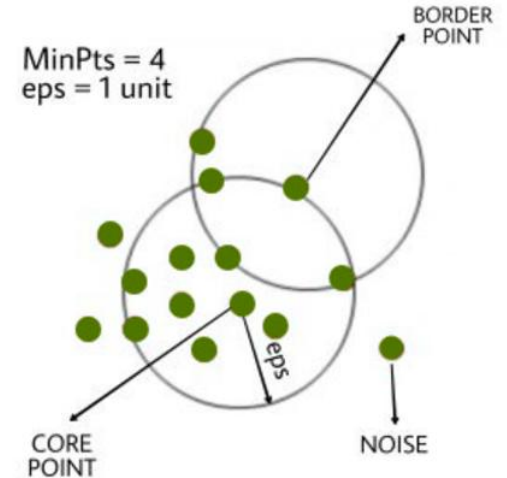
Clustering – K-Medoids

K-Medoids is a **clustering** algorithm similar to K-Means, but instead of using centroids (which are the averages of data points), it selects **actual data points**—called **medoids**—as the centers of clusters. This makes it more robust to noise and outliers.



Clustering – DBSCAN

Clusters are regions of high point density separated by areas of low density. **DBSCAN** identifies **clusters** and **noise** (outliers) based on this concept: for each point in a cluster, its surrounding area within a given radius must contain at least a **minimum number of neighboring points**.



Dataset

wine_quality_learning

- A **.csv file**: This file contains information about the **various properties of the** wine, such as acidity, alcohol, pH, etc. Specific for **training** the model.

winequality_test

- A **.csv file**: This file contains information about the **various properties of the** wine, such as acidity, alcohol, pH, etc. Specific for **testing** the model.

Both share the exact same features.

Dataset - Features

12 attributes:

- **fixed acidity**: Total amount of organic acids found in wine.
- **volatile acidity**: Quantity of acetic acid generated by yeast or bacteria.
- **citric acid**: The quantity of citric acid.
- **residual sugar**: The sugar remaining in wine after fermentation.
- **chlorides**: Amount of salt (chlorides) in the wine.
- **free sulfur dioxide**: Amount of free sulfur dioxide in the wine.
- **total sulfur dioxide**: Total amount of sulfur dioxide in the wine.
- **density**: Wine's density, related to the alcohol and sugar content.
- **pH**: Measures the wine's acidity or alkalinity. A lower pH indicates higher acidity.
- **sulphates**: Sulphates quantity, which act as a preservative and antioxidant.
- **alcohol**: Percentage of alcohol in the wine.
- **quality**: The quality rating of the wine, ranging from 0 (poor) to 10 (excellent).

Dataset - Features

- **fixed acidity:** This is the total amount of organic acids found in wine, primarily succinic, citric, malic, and tartaric acids. The wine's stability, color, and flavor are all influenced by these acids. Depending on the grape variety and climate, wine's average fixed acidity can range from 4 to 12 grams per liter (g/L).
- **volatile acidity:** This is the quantity of acetic acid generated by yeast or bacteria during age or fermentation. Acetic acid adds to the degree of sourness and pungency. It is optimal for wine to have an average volatile acidity level of less than 0.4 g/L. When volatile acidity above the sensory threshold of 0.6 to 0.9 g/L for red wines and 0.5 to 0.7 g/L for white wines, it is deemed a wine defect.
- **citric acid:** Adds a fresh, citrus-like acidity to the wine. Although present in small amounts, it contributes to the wine's overall acidity and liveliness. The quantity of citric acid is usually very low and needs to be measured in decimal points.

Dataset - Features

- **residual sugar:** Refers to the sugar remaining in wine after fermentation, affecting its sweetness. Dry wines have low residual sugar, while sweeter wines have higher levels.
- **chlorides:** Measures the amount of salt (chlorides) in the wine. High levels can negatively affect the flavor.
- **free sulfur dioxide:** Refers to the amount of free sulfur dioxide in the wine, which protects it from oxidation and microbial growth.
- **total sulfur dioxide:** Represents the total amount of sulfur dioxide in the wine, including both free and bound forms. High levels can affect flavor and safety.

Dataset - Features

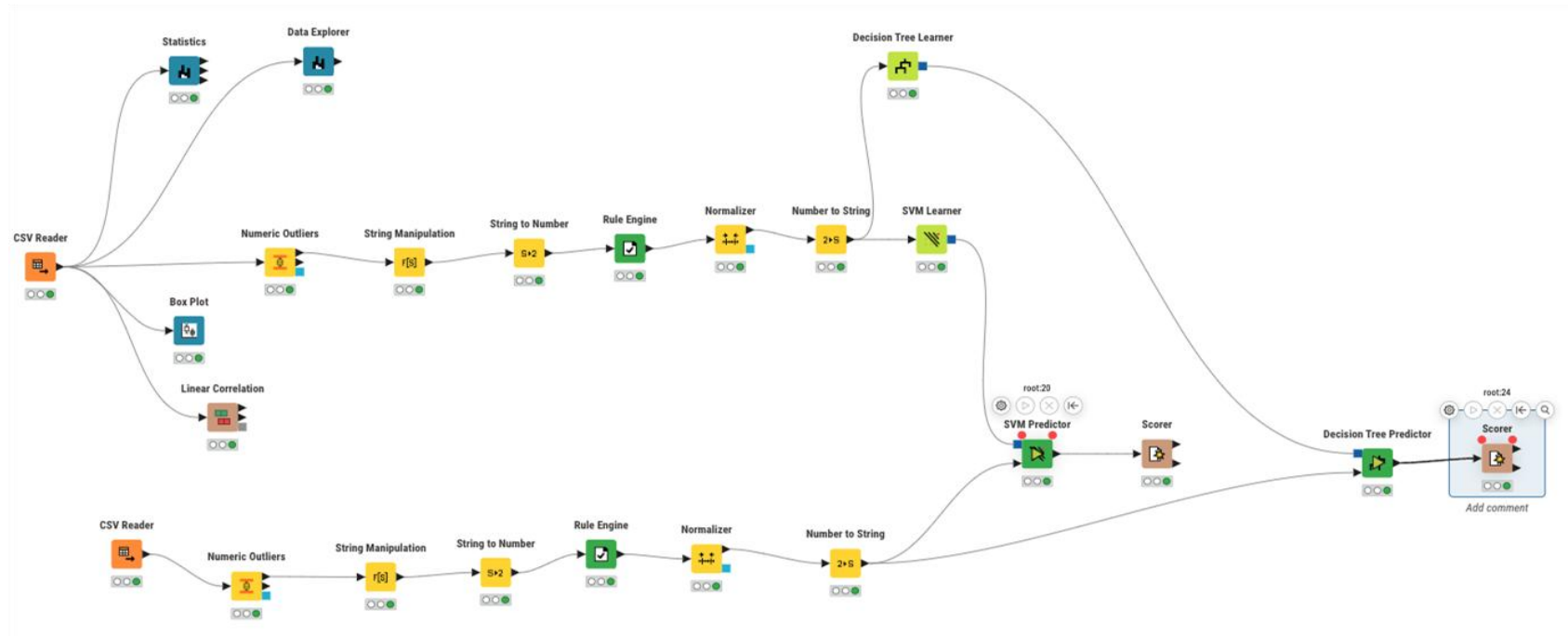
- **density:** Refers to the wine's density, which is related to the alcohol and sugar content. Density is measured as a continuous variable, often requiring decimals for accuracy. A double type handles this range of precise values effectively.
- **pH:** Measures the wine's acidity or alkalinity. A lower pH indicates higher acidity.
- **sulphates :** Refers to sulphates, which act as a preservative and antioxidant.

Dataset - Features

- **alcohol (double type):** Represents the percentage of alcohol in the wine, which influences its body and taste.
- **quality (string type):** The quality rating of the wine, usually based on sensory data, ranging from 0 (poor) to 10 (excellent).

Data preparation for **training** and **testing** must be **consistent**. This means that transformations applied to the training set must be applied **equally** to the test set.

Workflow



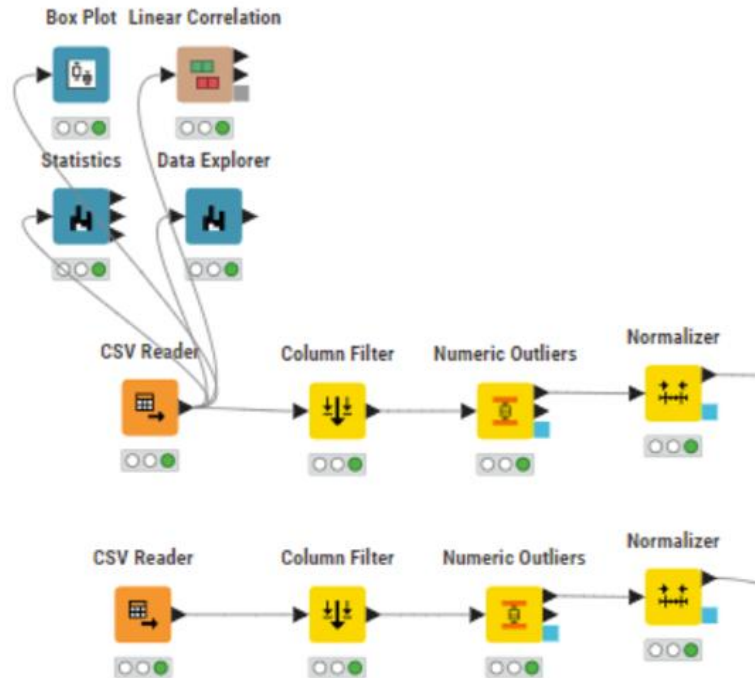
Goal: Build a K-Means model to cluster wines using dataset features. The objective is to uncover patterns in wine profiles and their relationship to quality ratings.

- **Create a new workflow**
 - Create a new workflow named **wine_clustering**.
 - Copy the pre-processing steps you previously applied to this dataset, as described below. **Copy only the nodes** between the **CSV Reader** node (inclusive) and the **Normalizer** node (inclusive).
 - **Paste** these nodes into the new **wine_clustering** workflow.
- **Remove the target**

Because this is an **unsupervised** task, the target variable **must be excluded**:

- Insert a **Column Filter** node **immediately after** the data reading step.
- Configure it to **exclude** the **quality** column.
- **Delete** any downstream nodes that processed the **quality** column, since it will no longer be part of the workflow.

Workflow



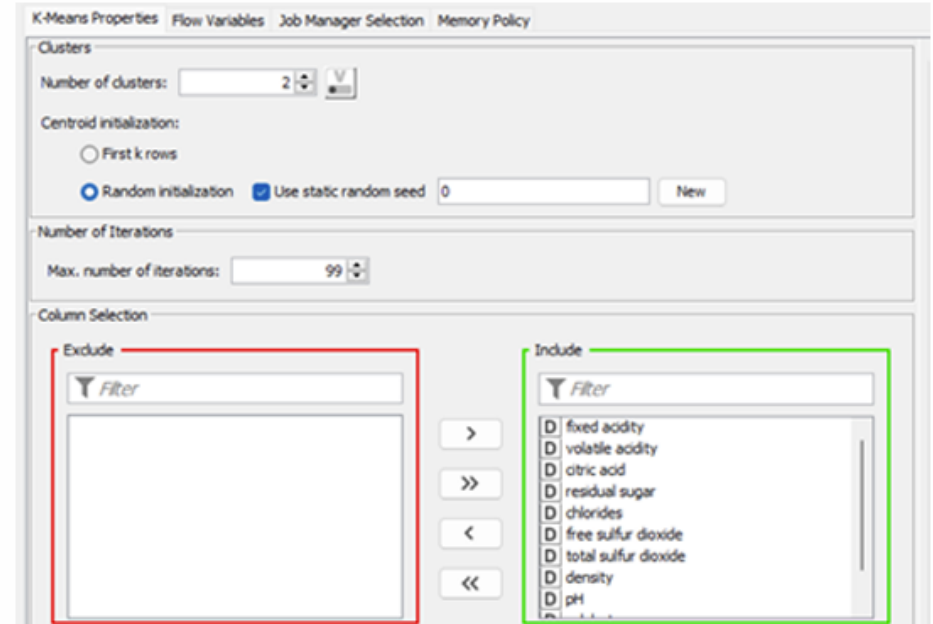


Train the Model

The **upper section of the workflow** contains the **training nodes**. Therefore, the corresponding node representing the model should be connected directly after these training nodes.

Connect the **K-Means node**.

Open the K-Means node's **configuration panel** and start with **2 clusters**.



The screenshot shows the 'K-Means Properties' configuration panel with the following settings:

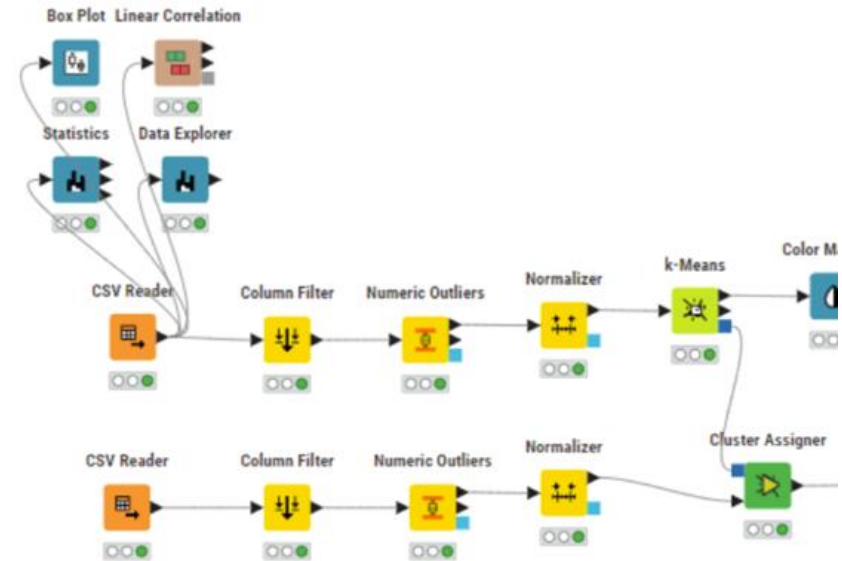
- Clusters:**
 - Number of clusters: 2
 - Centroid initialization: ☐ First k rows, ☒ Random initialization
 - ☒ Use static random seed: 0
 - New button
- Number of Iterations:**
 - Max. number of iterations: 99
- Column Selection:**
 - Exclude:** (Red box) Filter
 - Include:** (Green box) Filter
 - D fixed acidity
 - D volatile acidity
 - D citric acid
 - D residual sugar
 - D chlorides
 - D free sulfur dioxide
 - D total sulfur dioxide
 - D density
 - D pH
 - Navigation buttons: >, >>, <, <<



Cluster Assigner

Connect the **blue output port (model)** of the **K-Means node** to the **blue input port** of the **Cluster Assigner node**.

Connect the **gray output port (data)** of the **Normalizer node** to the **gray input port** of the **Cluster Assigner node**.

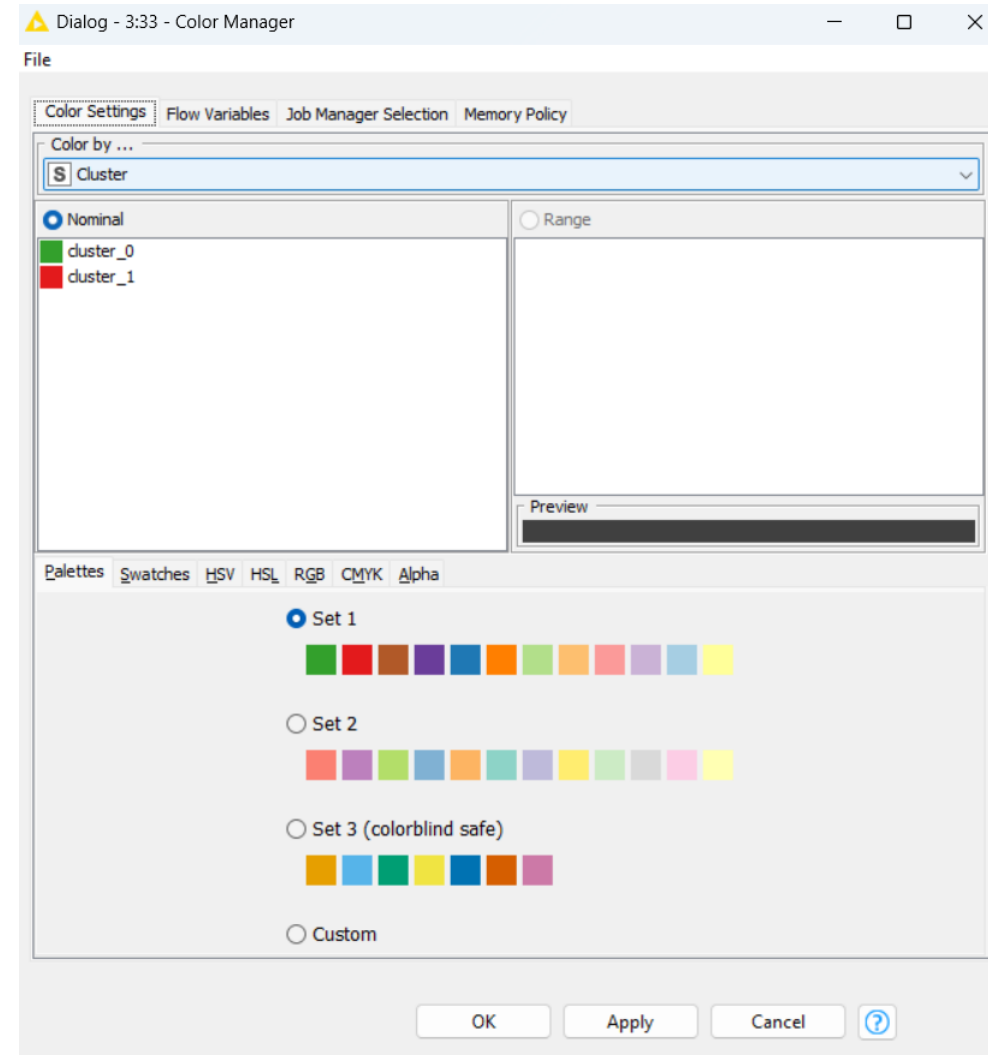


Color Manager

Connect the **output port** of the **K-Means** node to the **input port** of the **Color Manager** node.

Open the configuration panel.

Assign cluster colors.



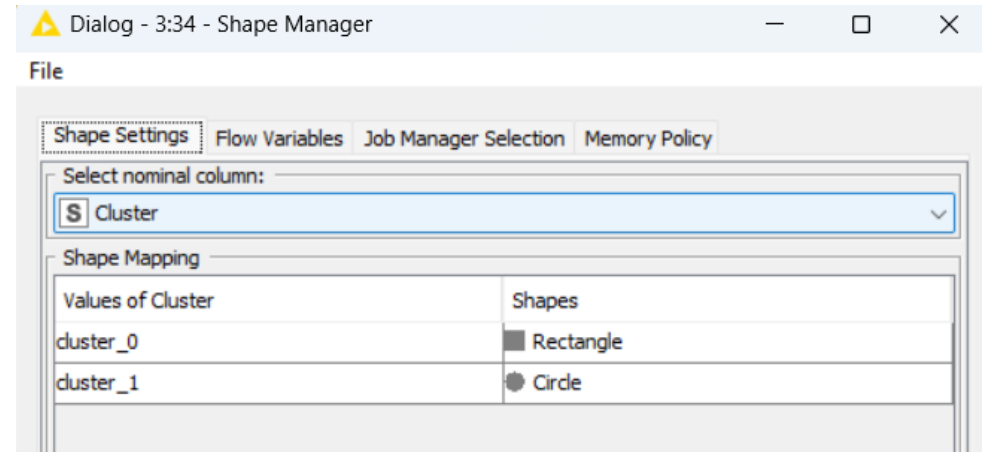


Shape Manager

Connect the **output port** of the **Color Manager** node to the **input port** of the **Shape Manager** node.

Open the configuration panel.

Assign desired shapes.





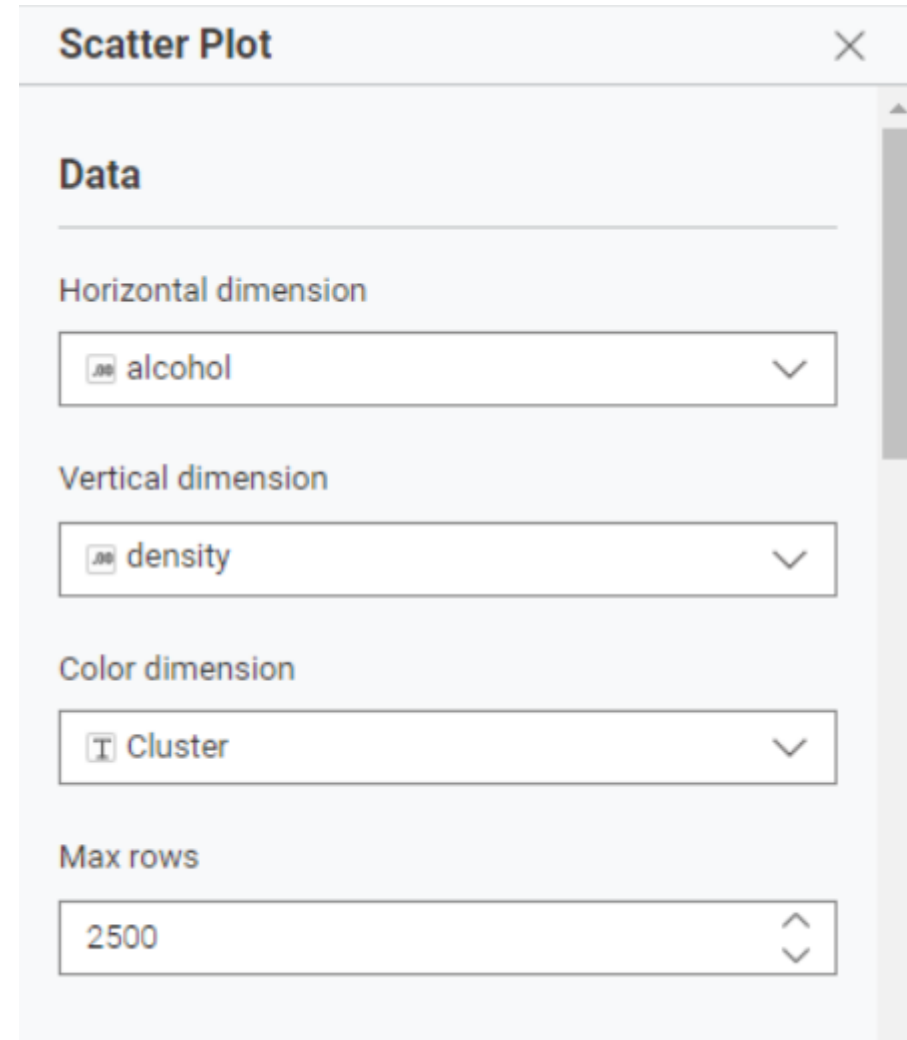
Scatter Plot

Connect the **output port** of the **Shape Manager** node to the **gray input port** of the **Scatter Plot**.

Open the Scatter Plot configuration.

Select two numerical columns for the X-axis and Y-axis (e.g., alcohol and volatile acidity).

Note: The Scatter Plot shows colors only, not shapes.

A screenshot of the Scatter Plot configuration window. The window has a title bar with a close button. The main area is titled "Data" and contains four configuration sections: "Horizontal dimension" with a dropdown menu showing "alcohol", "Vertical dimension" with a dropdown menu showing "density", "Color dimension" with a dropdown menu showing "Cluster", and "Max rows" with a text input field showing "2500". Each dropdown menu has a small icon on the left and a chevron on the right. The "Max rows" field has up and down arrows on the right.

Scatter Plot

Data

Horizontal dimension

alcohol

Vertical dimension

density

Color dimension

Cluster

Max rows

2500

Scatter Plot

Scatter Plot

Data

Horizontal dimension

Vertical dimension

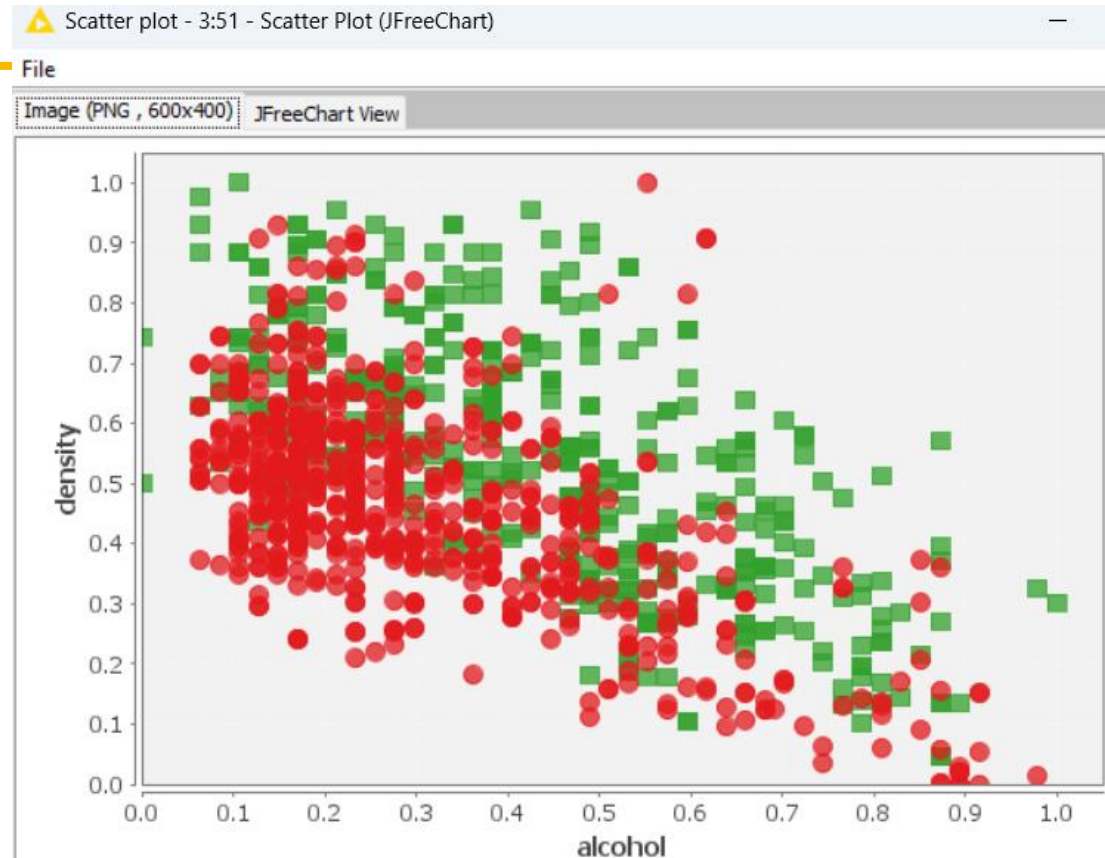
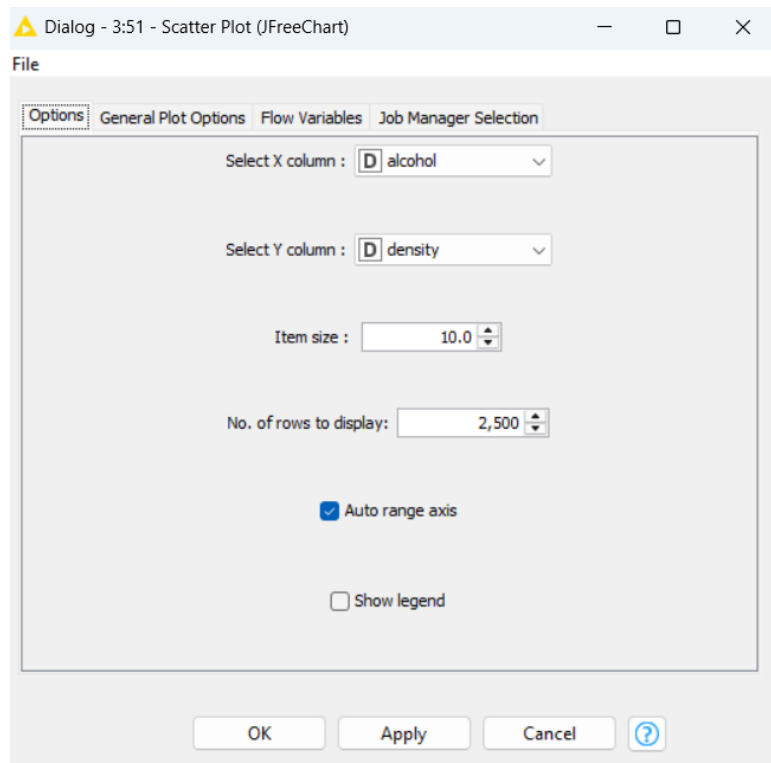
Color dimension

Max rows

Scatter Plot



Scatter Plot - JFreeChart



Evaluate the Model

Final Step: Evaluate the model

Measure:

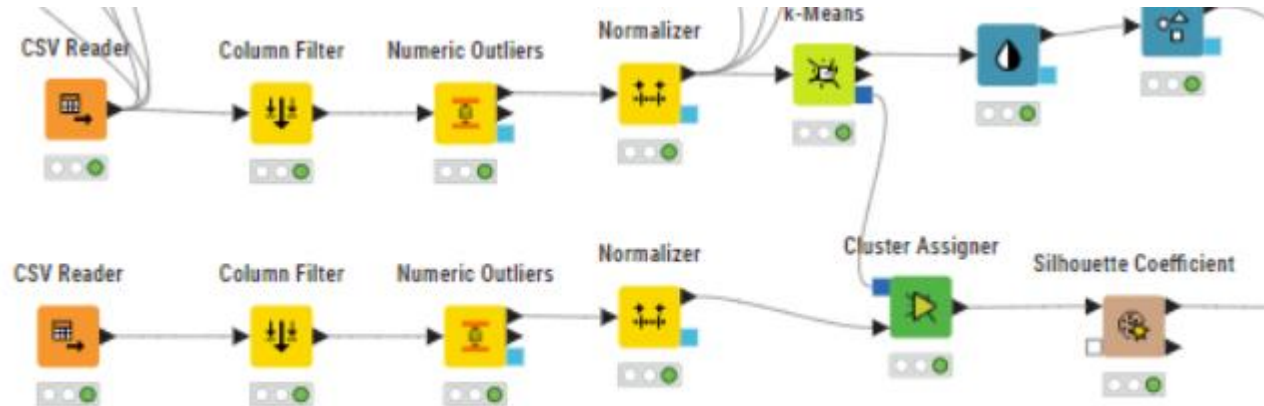
Silhouette Score



- Assesses a dataset's well-defined clusters. The cohesiveness and separation between clusters are quantified.
- Better-defined clusters are indicated by higher scores, which range from -1 to 1
- An object is **well-matched** to its own cluster if its score is **close to 1**.
- A score of **about -1** suggests that the object might be in the **incorrect cluster**.

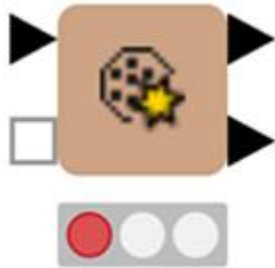
Silhouette Coefficient

Connect the **output port** of the **Cluster Assigner** node to the **input port** of the **Silhouette Coefficient** node.



Silhouette Coefficient

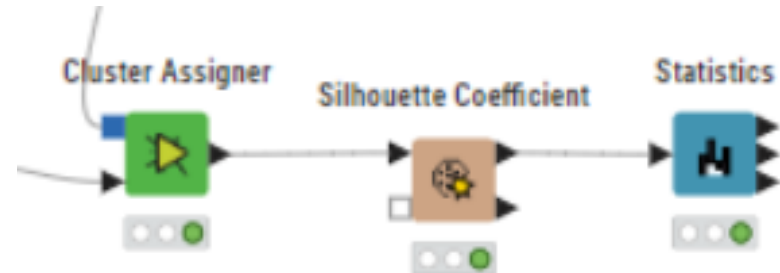
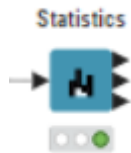
Final Step: Evaluate the model
Node: Silhouette Coefficient



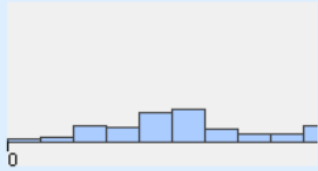
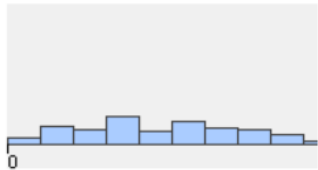
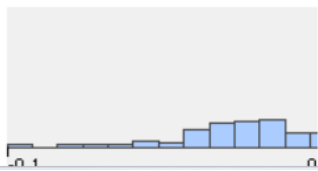
Cluster <i>String</i>	Silhouett... <i>Number (dou...</i>
cluster_1	0.261
cluster_1	0.237
cluster_1	0.109
cluster_0	-0.02
cluster_1	0.342
cluster_1	0.337

Statistics

Connect the **gray output port** of the **Silhouette Coefficient** node to the **gray input port** of the **Statistics** node.



Statistics

sulphates	0.0	0.5325	?	1	0.2296	0.3346	-0.5197	0	0	0	
alcohol	0.0	0.4527	?	1	0.2341	0.167	-0.7838	0	0	0	
Silhouette Coefficient	-0.1323	0.2239	?	0.3727	0.0966	-1.1152	1.9246	0	0	0	

Statistics

Evaluate the Model

- **Maximum score = 0.3727.** Positive value, some data points are relatively well-clustered, with a moderate degree of separation from other clusters.
- **Minimum score = -0.1323.** Some data points are **misclassified**, meaning they are closer to points in other clusters than their own.

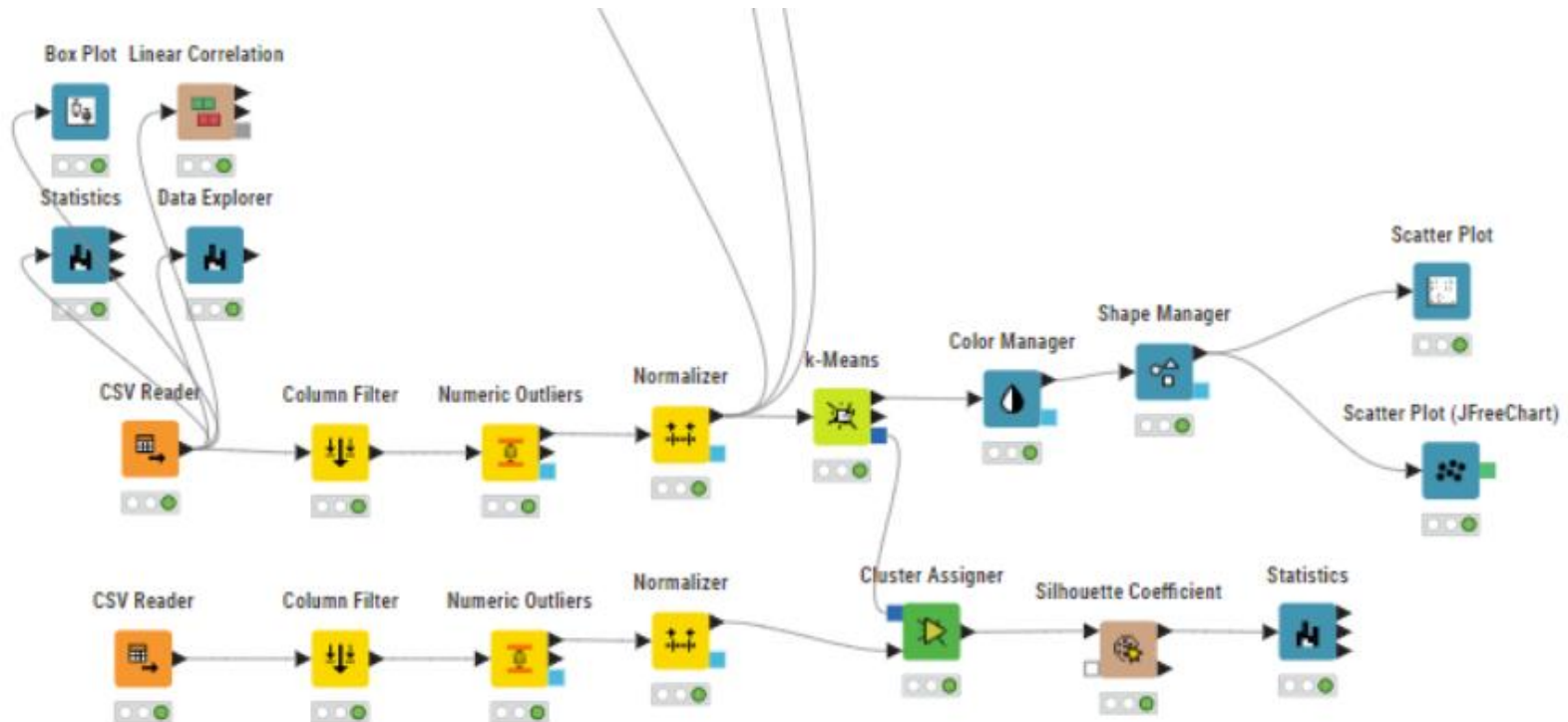
Silhouette Coefficient	-0,1323	0,2239	?	0,3727
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Statistics

Evaluate the Model

- The presence of both positive and slightly negative scores suggests that, while the clustering model generally performs well for many points, there are some points for which the model **struggles to clearly distinguish**.
- Could be related to **overlapping cluster boundaries** or **complex patterns** in the data.
- Further tuning might be necessary for improved results.
- Experiment **different values for k** and assess the final results.

Workflow



TASK 2

Goal:

- Understand **K-Medoids**
- Develop a **K-Medoids model** to effectively **group wines** based on relevant features
- Analyze the **performance** of the final model



K-Medoids

- Similar to k-means, but instead of using centroids (averages of data points), it selects **actual data points (medoids)**;
- Medoids represent the “center” for each cluster;
- **More robust** to noise and outliers than K-Means;

K-Medoids

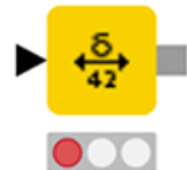
- **Initialization:** The algorithm randomly selects k data points as the initial medoids;
- **Assignment:** The algorithm assigns each data point to the nearest medoid based on a chosen distance metric;
- **Update:** For each cluster, the algorithm finds a new medoid that minimizes the total distance between the medoid and all points in the cluster;
- **Iterate:** The assignment is repeated and such as the update steps until the medoids no longer change.

K-Medoids

Exercise:

First Step: Compute distances

Node: Numeric Distances



Calculates distances between numeric features for clustering

Distance Configuration | Flow Variables | Job Manager Selection

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

No columns in this list

☒ Enforce exclusion

Include

Filter

- ☒ fixed acidity
- ☒ volatile acidity
- ☒ citric acid
- ☒ residual sugar
- ☒ chlorides
- ☒ free sulfur dioxide
- ☒ total sulfur dioxide
- ☒ density

☐ Enforce inclusion

Distance Selection

☒ Standard Distance (Euclidean/Manhattan)

Configuration

☒ Euclidean

☐ Manhattan

☐ Maximum

☐ Custom 'p'

☐ Normalize distance (Requires normalized input vectors)

Missing Values

☒ Fail (fails if a missing value cell occurs during computation.)

K-Medoids

Exercise:

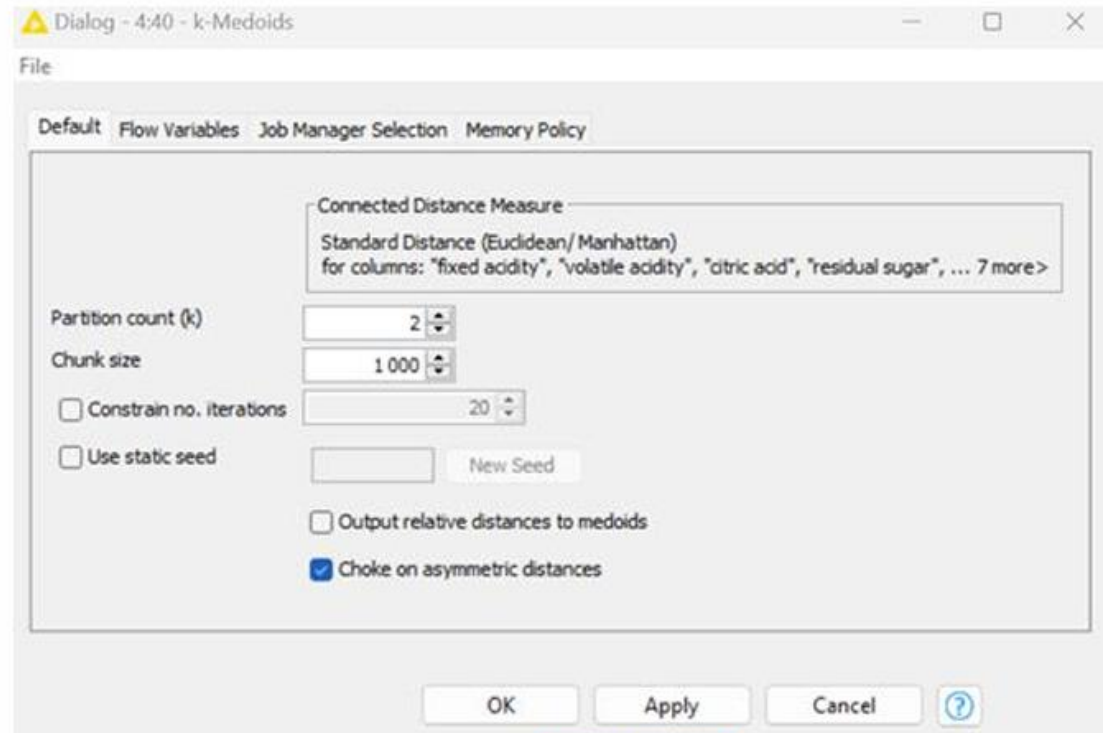
Second Step: Train the model

Node: K-Medoids



Trains a K-Medoids model

Initial number of clusters: 2



K-Medoids

Exercise:

Third Step

Node: Cluster Assigner



- Assigns new data points to the **nearest cluster** based on an existing clustering model;
- Allows users to **classify data** according to predefined clusters.

K-Medoids

Exercise:

Fourth Step

Node: Color Manager



*Allows users to assign
specific colors to data
values.*



K-Medoids

Exercise:

Fifth Step

Node: Shape Manager



*Allows users to assign
specific shapes to data
values.*

Shape Settings | Flow Variables | Job Manager Selection | Memory Policy

Select nominal column:

Shape Mapping

Values of Cluster	Shapes
cluster_0	 Rectangle
cluster_1	 Circle

K-Medoids

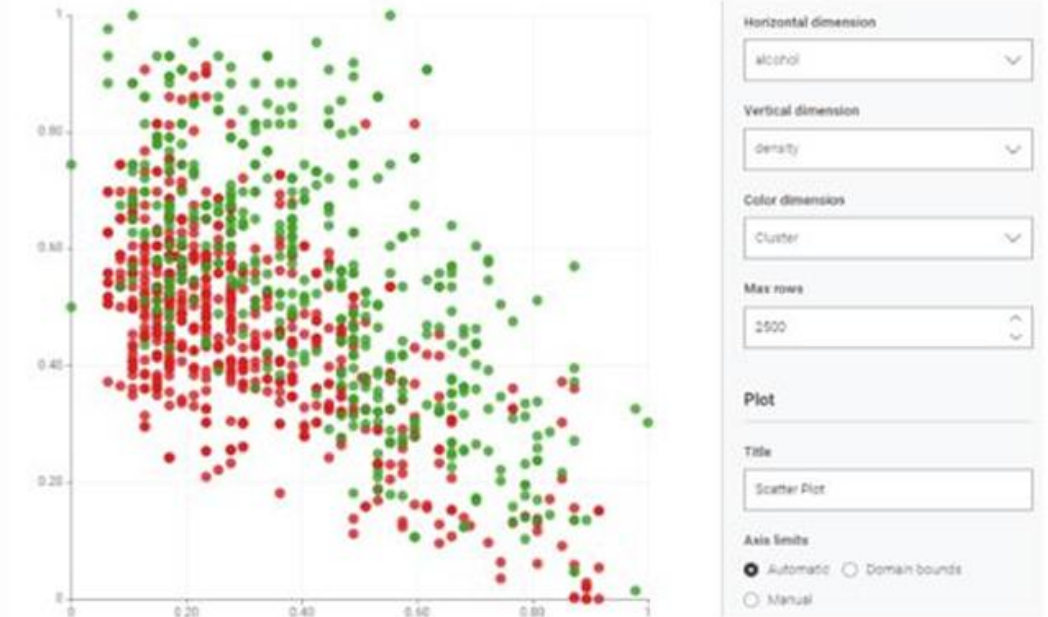
Exercise:

Sixth Step: Visualization

Node: Scatter Plot



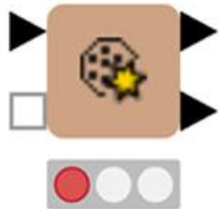
*Visualizes clusters by
representing data
points*



K-Medoids

Exercise:

Final Step: Evaluate the model
Node: Silhouette Coefficient



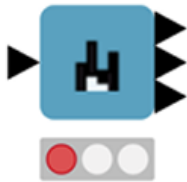
alcohol <i>Number (dou...</i>	Cluster <i>String</i>	Silhouett... <i>Number (dou...</i>
0.149	Row85	0.373
0.234	Row85	0.261
0.234	Row85	0.32
0.234	Row1153	0.266
0.149	Row85	0.373
0.149	Row85	0.375

K-Medoids

Exercise:

Final Step: Evaluate the model

Node: Statistics



sulphates	0.0	0,4636	?	1
alcohol	0.0	0,3497	?	1
Silhouette Coefficient	-0,1635	0,1917	?	0,4013

K-Medoids

Evaluate the Model

- **Maximum score = 0.4013.** Positive value, moderate separation between clusters, meaning that some of the clusters are relatively well-formed.
- **Minimum score = -0.1635.** Some data points are **misclassified**, meaning they are closer to points in other clusters than their own.

Silhouette Coefficient	-0,1635	0,1917	?	0,4013
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K-Medoids

Evaluate the Model

- The presence of both positive and slightly negative scores suggests that, while the clustering model generally performs well for many points, there are some points for which the model **struggles to clearly distinguish**.
- The results demonstrate the value of the K-Medoids algorithm in producing meaningful clusters, when compared to K-Means.
- However, the presence of misclassified points suggests that further refinement may be necessary.

Final Workflow



CHALLENGE



KNIME Analytics Platform

Using the same **Wine Quality** workflow, try applying the **DBScan** algorithm.

- Follow the same process applied for the K-Medoids algorithm. Start with an **epsilon of 0.4** and **min points of 6**.

Remember:

- Typical **Epsilon** values: range of **0.1 to 0.5**. Lower values: result in more defined clusters, but with many points classified as noise. Larger values: may result in few large clusters.
- Typical **min points** value: **5 to 10**. Lower values: DBSCAN may generate many small clusters. Larger values: the algorithm may not identify distinct clusters.



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